

Electric Circuits II Project

Report

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Tutorial Number

27

Theoretical background:

The chosen circuit is an **Inverting Active Band-Pass Filter**. It consists of an operational amplifier with a series resistor-capacitor network at the input (R_1 and C_1) and a parallel resistor-capacitor network in the feedback loop (R_2 and C_2).

Gain Equation Derivation (Phasor Domain)

Assuming the non-inverting terminal (V_p) is connected to ground, making $V_p = 0$. Since the operational amplifier used is ideal, then $V_n = V_p = 0$.

Applying Nodal Analysis at node V_n :

$$\frac{V_n - V_{in}}{Z_{in}} + \frac{V_n - V_{out}}{Z_f} = 0 \quad (1)$$

Expanding the impedances Z_{in} (series R_1, C_1) and Z_f (parallel R_2, C_2):

$$\frac{V_n - V_{in}}{R_1 + \frac{1}{j\omega C_1}} + \frac{V_n - V_{out}}{\left(\frac{1}{R_2} + j\omega C_2\right)^{-1}} = 0 \quad (2)$$

Simplifying the impedance fractions:

$$\frac{V_n - V_{in}}{\frac{1+j\omega R_1 C_1}{j\omega C_1}} + \frac{V_n - V_{out}}{\frac{R_2}{1+j\omega R_2 C_2}} = 0 \quad (3)$$

Substituting $V_n = 0$:

$$\frac{-V_{in}}{\frac{1+j\omega R_1 C_1}{j\omega C_1}} + \frac{-V_{out}}{\frac{R_2}{1+j\omega R_2 C_2}} = 0 \quad (4)$$

Rearranging to solve for the voltage gain $H(j\omega) = \frac{V_{out}}{V_{in}}$:

$$\frac{V_{out}}{\frac{R_2}{1+j\omega R_2 C_2}} = \frac{-V_{in}}{\frac{1+j\omega R_1 C_1}{j\omega C_1}} \implies \frac{V_{out}}{V_{in}} = -\frac{\frac{R_2}{1+j\omega R_2 C_2}}{\frac{1+j\omega R_1 C_1}{j\omega C_1}} \quad (5)$$

$$H(j\omega) = \frac{-j\omega R_2 C_1}{(1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2)} \quad (6)$$

Expanding the denominator and separating real and imaginary parts:

$$H(j\omega) = \frac{-j\omega R_2 C_1}{(1 - \omega^2 R_1 C_1 R_2 C_2) + j\omega(R_1 C_1 + R_2 C_2)} \quad (7)$$

Gain Magnitude and Resonant Frequency (f_0)

The magnitude of the voltage gain $|H_v(j\omega)|$ is obtained by dividing the magnitude of the numerator by the magnitude of the denominator:

$$|H(j\omega)| = \frac{\omega R_2 C_1}{\sqrt{(1 - \omega^2 R_1 C_1 R_2 C_2)^2 + \omega^2 (R_1 C_1 + R_2 C_2)^2}} \quad (8)$$

To find the resonant frequency (ω_0) where the gain is maximum, we can minimize the square of the inverse of the magnitude function. Let $u(\omega) = \frac{1}{|H(j\omega)|^2}$:

$$u(\omega) = \frac{1 - 2\omega^2 R_1 C_1 R_2 C_2 + \omega^4 (R_1 C_1 R_2 C_2)^2 + \omega^2 (R_1 C_1 + R_2 C_2)^2}{\omega^2 R_2^2 C_1^2} \quad (9)$$

Simplifying the expression into separate terms:

$$u(\omega) = \frac{1}{\omega^2 R_2^2 C_1^2} + \frac{\omega^2 (R_1 C_1 R_2 C_2)^2}{R_2^2 C_1^2} + \text{Constant Terms} \quad (10)$$

Differentiating $u(\omega)$ with respect to ω and equating to zero to find the minimum:

$$\frac{du}{d\omega} = -\frac{2}{\omega^3 R_2^2 C_1^2} + \frac{2\omega (R_1 C_1 R_2 C_2)^2}{R_2^2 C_1^2} = 0 \quad (11)$$

$$\frac{1}{\omega^3} = \omega (R_1 C_1 R_2 C_2)^2 \implies \omega_0^4 = \frac{1}{(R_1 C_1 R_2 C_2)^2} \quad (12)$$

$$\omega_0 = \frac{1}{\sqrt{R_1 C_1 R_2 C_2}} \implies f_0 = \frac{1}{2\pi \sqrt{R_1 C_1 R_2 C_2}} \quad (13)$$

Filter design:

Based on the theoretical equations, the following component values were selected to achieve a target maximum gain between 1 and 10, operating in the kilohertz range for the hardware implementation.

Component Values

- Input Resistance (R_1): 1 k Ω
- Feedback Resistance (R_2): 5.6 k Ω
- Input Capacitance (C_1): 68 nF
- Feedback Capacitance (C_2): 5.6 nF

Calculated Parameters

Using the selected component values, the expected theoretical parameters for the simulation and hardware implementation are as follows:

1. Lower Cut-off Frequency (f_L):

$$f_L = \frac{1}{2\pi(1000)(68 \times 10^{-9})} \approx 2.34 \text{ kHz} \quad (14)$$

2. Upper Cut-off Frequency (f_H):

$$f_H = \frac{1}{2\pi(5600)(5.6 \times 10^{-9})} \approx 5.08 \text{ kHz} \quad (15)$$

3. Center/Resonant Frequency (f_0):

$$f_0 = \frac{1}{2\pi\sqrt{R_1C_1R_2C_2}} = \frac{1}{2\pi\sqrt{(1000)(68 \times 10^{-9})(5600)(5.6 \times 10^{-9})}} \approx 3.45 \text{ kHz} \quad (16)$$

4. Maximum Expected Gain ($|_{\max}|$): Substituting ω_0 into the magnitude equation yields the maximum gain:

$$\omega_0 = \frac{1}{\sqrt{R_1C_1R_2C_2}} \implies \omega_0^2 = \frac{1}{R_1C_1R_2C_2}$$

$$|H(j\omega_0)| = \frac{\omega_0 R_2 C_1}{\sqrt{(1 - \omega_0^2 R_1 C_1 R_2 C_2)^2 + \omega_0^2 (R_1 C_1 + R_2 C_2)^2}}$$

Substituting ω_0^2 inside the real part of the denominator:

$$|H(j\omega_0)| = \frac{\omega_0 R_2 C_1}{\sqrt{\left(1 - \omega_0^2 \left(\frac{1}{\omega_0^2}\right)\right)^2 + \omega_0^2 (R_1 C_1 + R_2 C_2)^2}}$$

$$|H(j\omega_0)| = \frac{\omega_0 R_2 C_1}{\sqrt{(1 - 1)^2 + \omega_0^2 (R_1 C_1 + R_2 C_2)^2}}$$

$$|H(j\omega_0)| = \frac{\omega_0 R_2 C_1}{\omega_0 (R_1 C_1 + R_2 C_2)}$$

$$|H_{\max}| = \frac{R_2 C_1}{R_1 C_1 + R_2 C_2} \quad (17)$$

$$|H_{\max}| = \frac{(5600)(68 \times 10^{-9})}{(1000)(68 \times 10^{-9}) + (5600)(5.6 \times 10^{-9})} \approx 3.83 \quad (18)$$

5. Expected Gain at Cut-off Frequencies ($|H_{\omega_L}|$ and $|H_{\omega_H}|$): The lower cut-off frequency is dictated by the input network: $\omega_L = \frac{1}{R_1 C_1}$. Substituting ω_L into the gain magnitude equation:

$$|H(j\omega_L)| = \frac{\left(\frac{1}{R_1 C_1}\right) R_2 C_1}{\sqrt{\left(1 - \left(\frac{1}{R_1 C_1}\right)^2 R_1 C_1 R_2 C_2\right)^2 + \left(\frac{1}{R_1 C_1}\right)^2 (R_1 C_1 + R_2 C_2)^2}}$$

$$|H(j\omega_L)| = \frac{\frac{R_2}{R_1}}{\sqrt{\left(1 - \frac{R_2 C_2}{R_1 C_1}\right)^2 + \left(1 + \frac{R_2 C_2}{R_1 C_1}\right)^2}}$$

Expanding the binomials inside the square root ($1 - 2x + x^2 + 1 + 2x + x^2 = 2 + 2x^2$):

$$|H(j\omega_L)| = \frac{\frac{R_2}{R_1}}{\sqrt{2 + 2\left(\frac{R_2 C_2}{R_1 C_1}\right)^2}} = \frac{R_2 C_1}{\sqrt{2((R_1 C_1)^2 + (R_2 C_2)^2)}}$$

Similarly, substituting the upper cut-off frequency $\omega_H = \frac{1}{R_2 C_2}$ yields an identical simplified expression due to the mathematical symmetry of the poles:

$$|H(j\omega_H)| = \frac{\frac{C_1}{C_2}}{\sqrt{\left(1 - \frac{R_1 C_1}{R_2 C_2}\right)^2 + \left(\frac{R_1 C_1}{R_2 C_2} + 1\right)^2}} = \frac{R_2 C_1}{\sqrt{2((R_1 C_1)^2 + (R_2 C_2)^2)}}$$

Therefore, the exact theoretical gain at both cut-off frequencies is identical. Substituting the component values:

$$|H_{\omega_L}| = |H_{\omega_H}| = \frac{(5600)(68 \times 10^{-9})}{\sqrt{2((1000 \cdot 68 \times 10^{-9})^2 + (5600 \cdot 5.6 \times 10^{-9})^2)}} \approx 3.60 \quad (19)$$

Circuit simulation:

The band-pass filter circuit was constructed using PSpice simulation software. The following figures detail the schematic diagram and the simulation profiles used for both AC sweep and transient (time-domain) analysis.

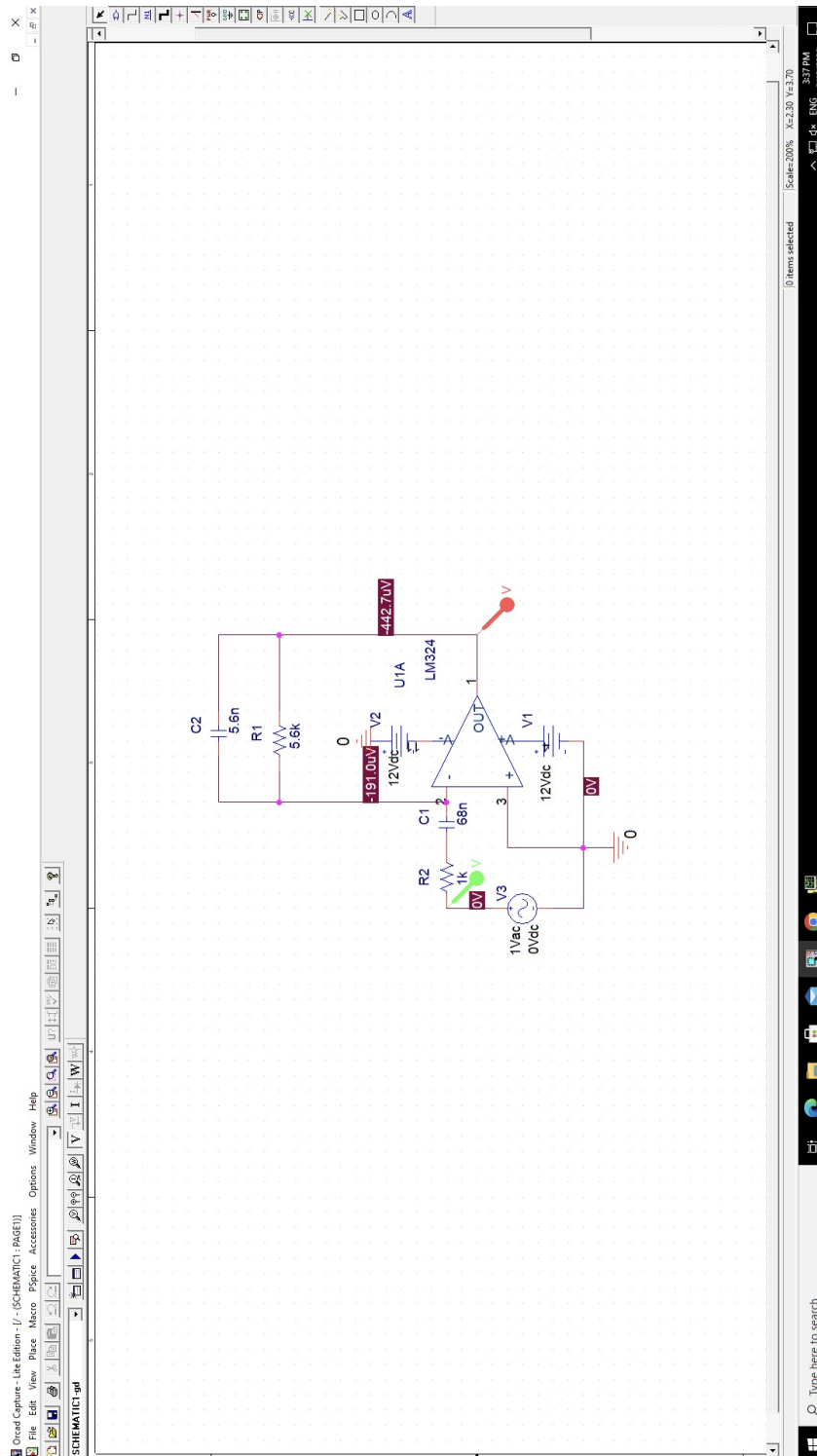


Figure 1: PSpice schematic diagram of the Inverting Band-Pass Filter.

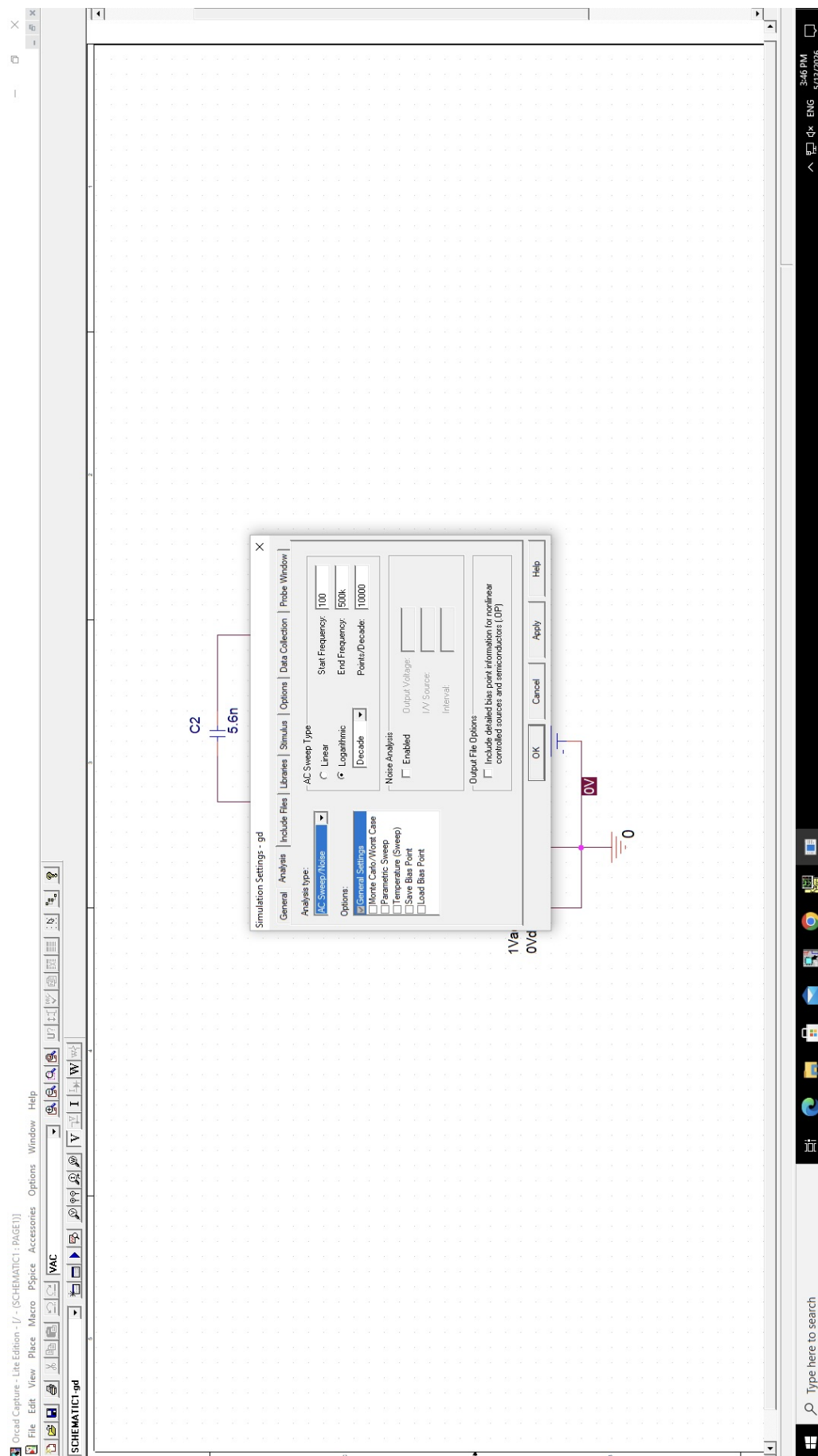


Figure 2: AC Sweep simulation profile settings as mentioned in the project description.

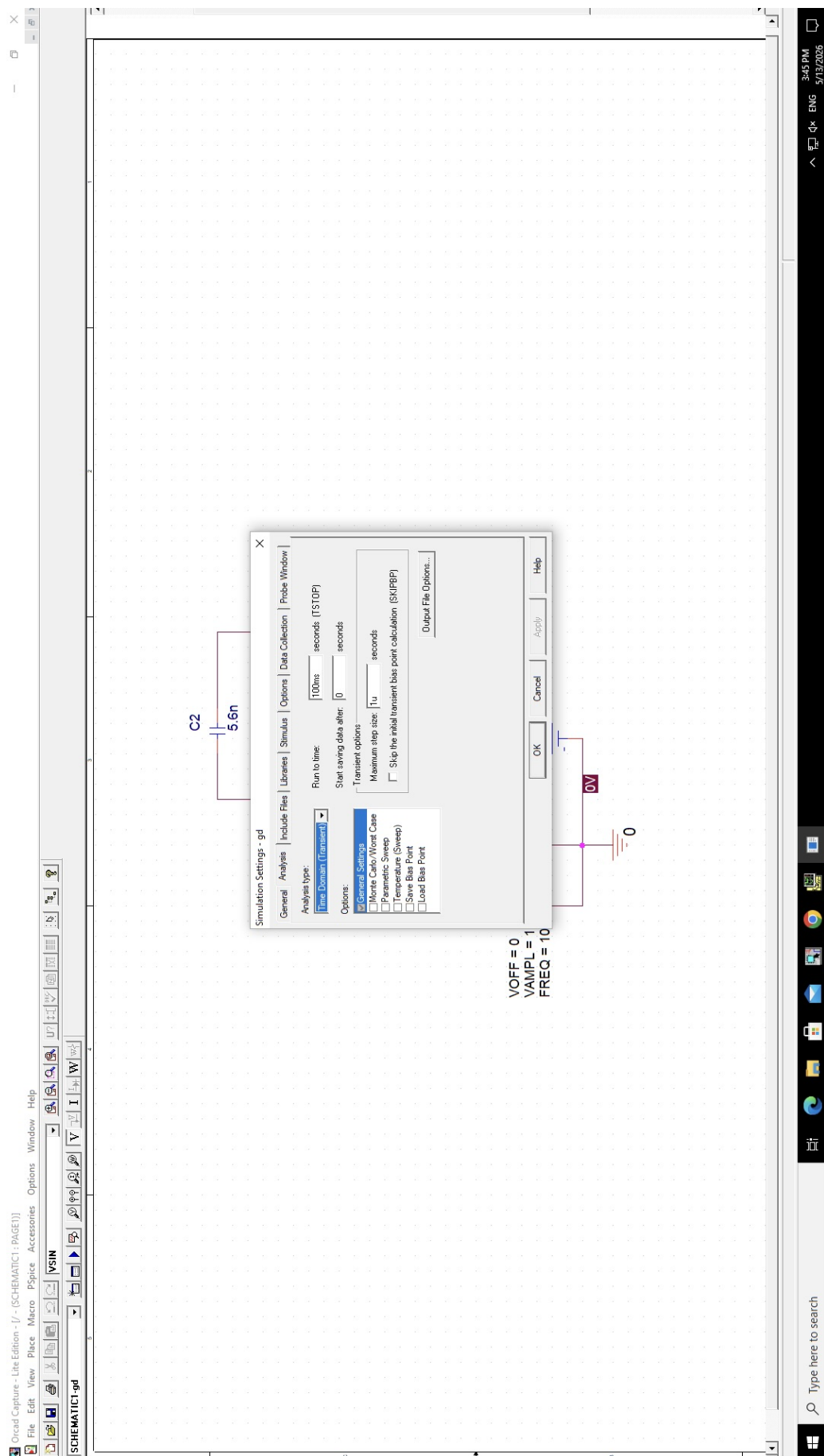


Figure 3: Time domain simulation profile settings as mentioned in the project description.

Simulation results:

i. AC Sweep Simulation

An AC sweep was performed to sketch the voltage gain magnitude versus frequency. As expected for a band-pass filter, the gain peaks at the center frequency and rolls off at higher and lower frequencies. The maximum gain is verified to be approximately 3.8 as calculated.

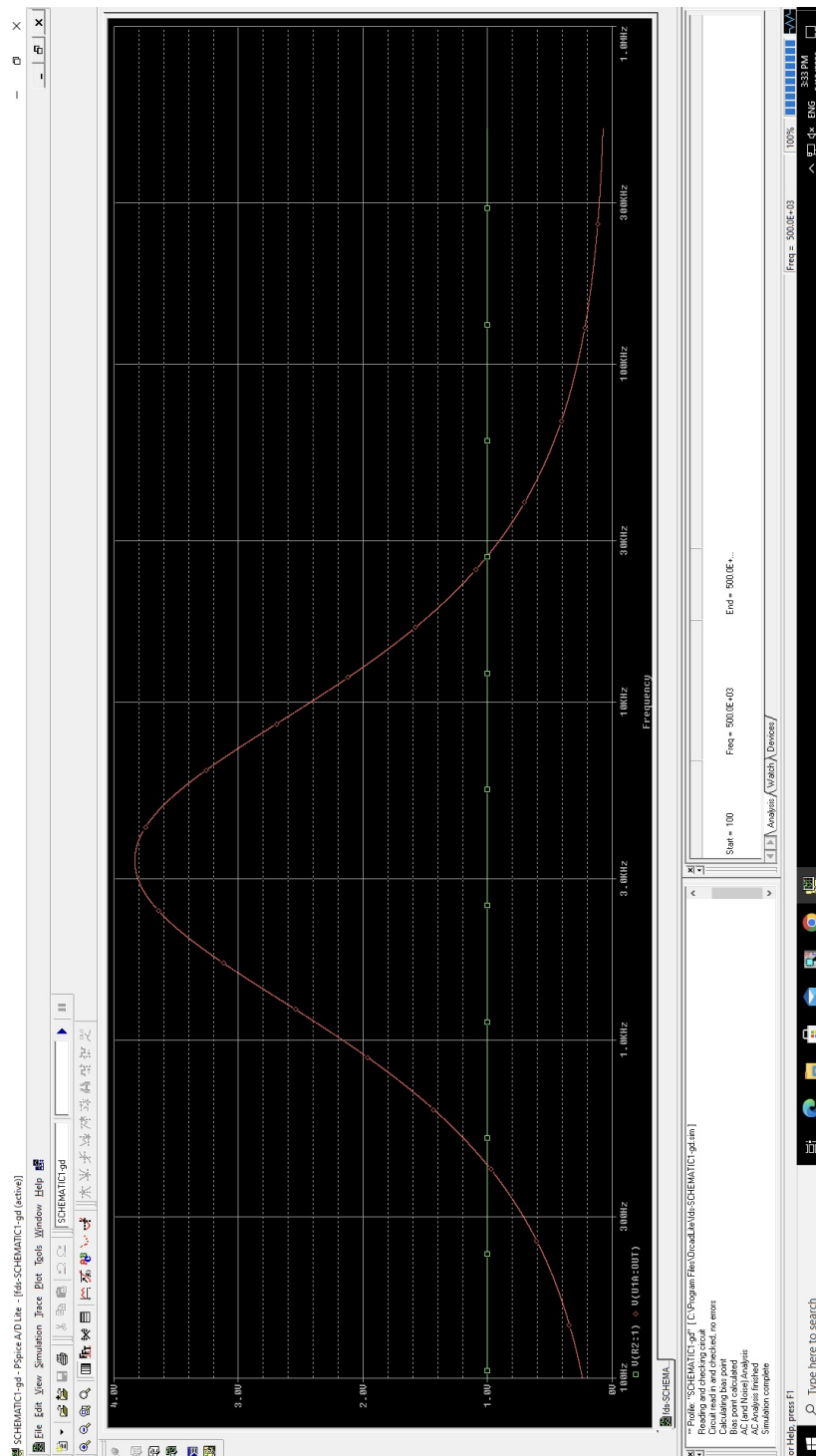


Figure 4: AC Sweep Simulation showing gain versus frequency.

ii. Time Domain Simulation

A time-domain analysis was conducted using a sinusoidal input voltage (V_{sin}) to observe the time-domain response. The output and input voltage amplitudes were compared at three different frequencies to demonstrate the band-pass attenuation characteristics.

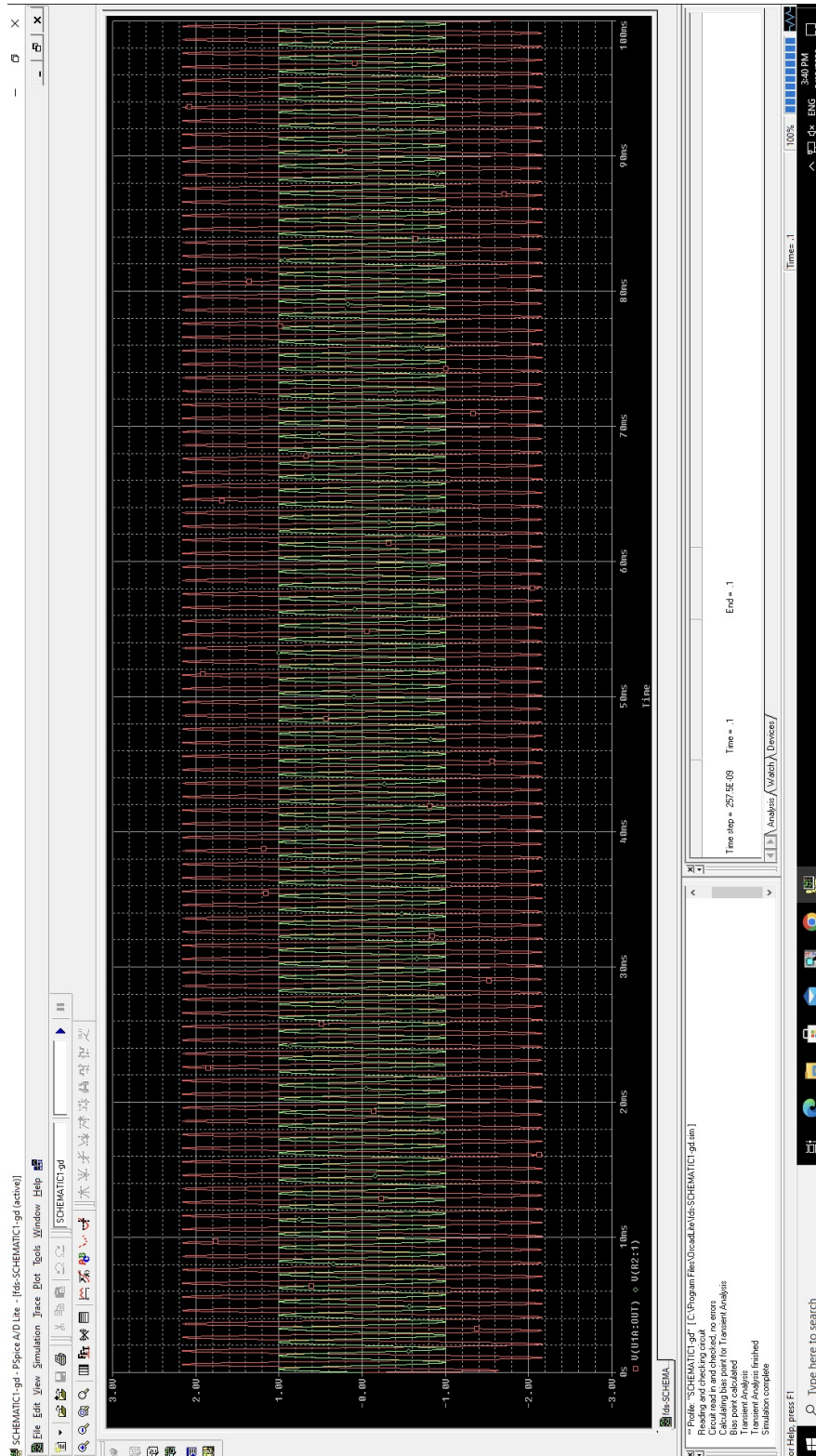


Figure 5: Time domain simulation at $f < f_0$ ($f = 1$ kHz). The gain here is approximately 2.2 (< 3.8) as expected from the AC sweep graph.

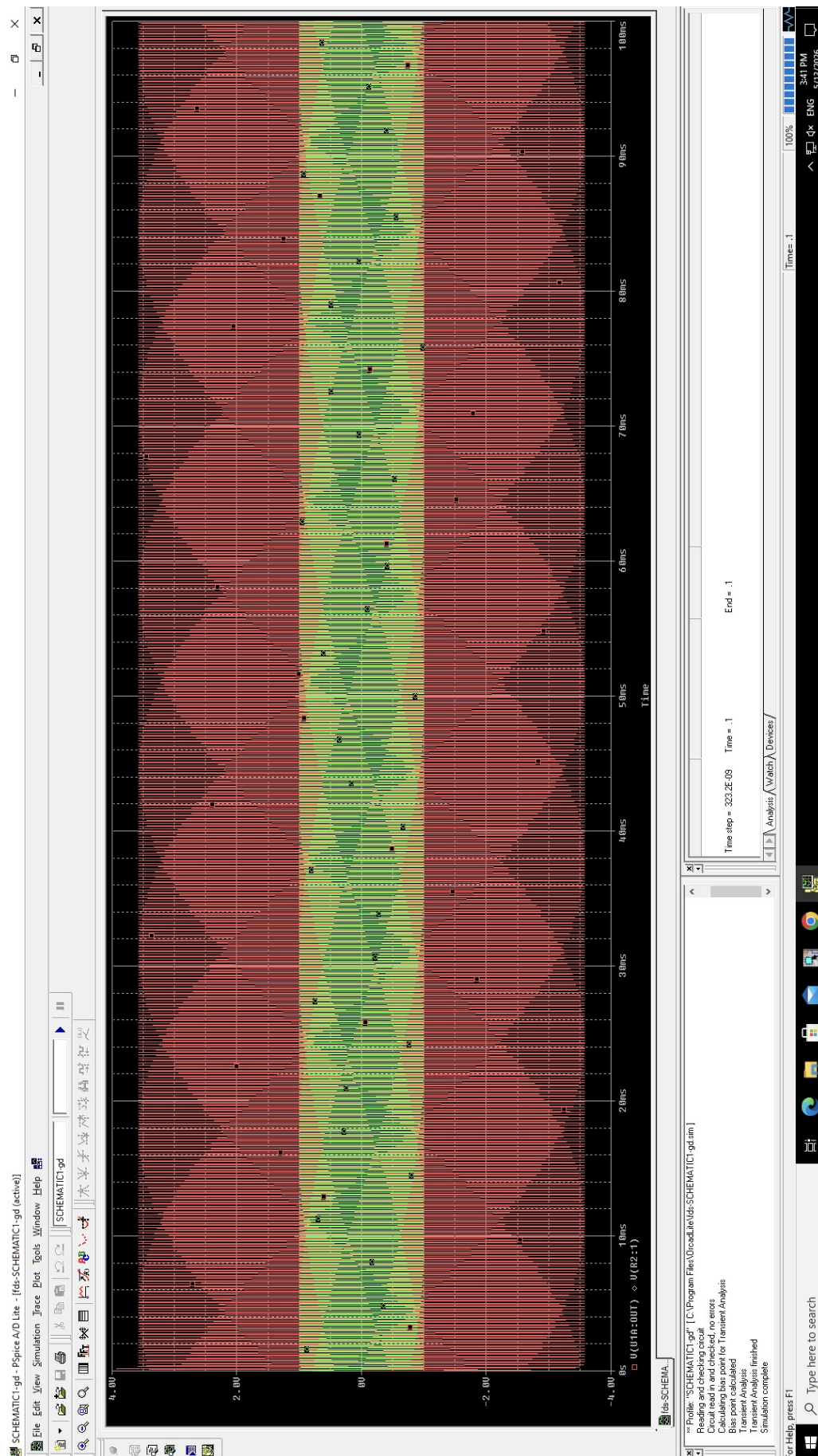


Figure 6: Time domain simulation at the resonant frequency $f = f_0 = 3.45$ kHz. The gain here is approximately 3.8 as expected from the calculations.

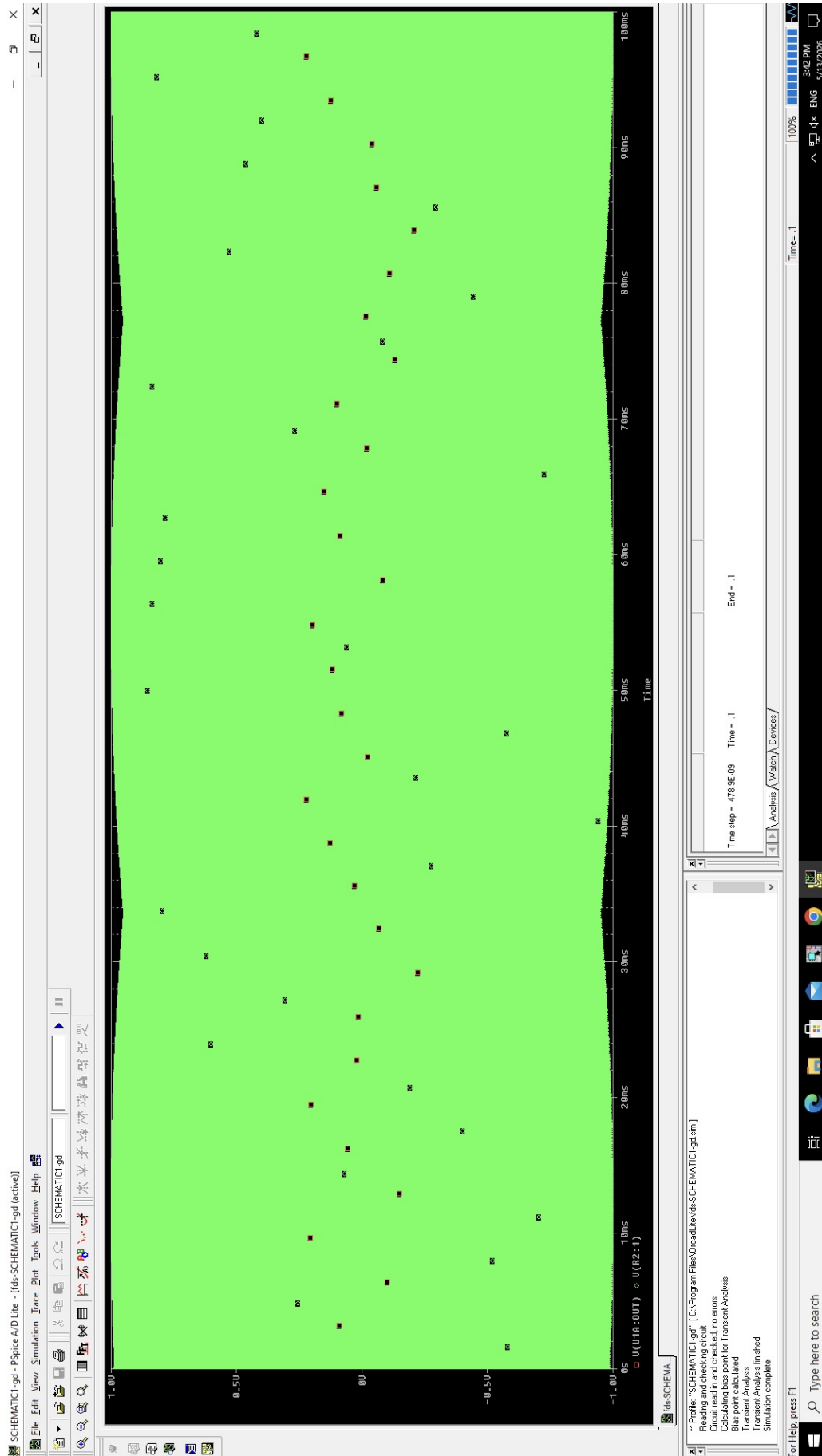


Figure 7: Time domain simulation at $f > f_0$ ($f = 100$ kHz). We notice here that the red points (V_{out}) are below the green ones (V_{in}) meaning the gain is < 1 as expected from the AC sweep graph.

Hardware implementation:

The band-pass filter circuit was physically constructed on a standard breadboard for testing and validation. The details of the purchased components used for the hardware implementation are listed below, followed by photographs of the assembled circuit.

Purchased Components

Component	Nominal Value / Part Number	Quantity
Resistor (R_1)	1 k Ω	1
Resistor (R_2)	5.6 k Ω	1
Capacitor (C_1)	68 nF	1
Capacitor (C_2)	5.6 nF	1
Operational Amplifier	LM741	1

Table 1: List of components purchased for the hardware implementation.

Hardware Pictures

The physical circuit was wired according to the theoretical schematic.

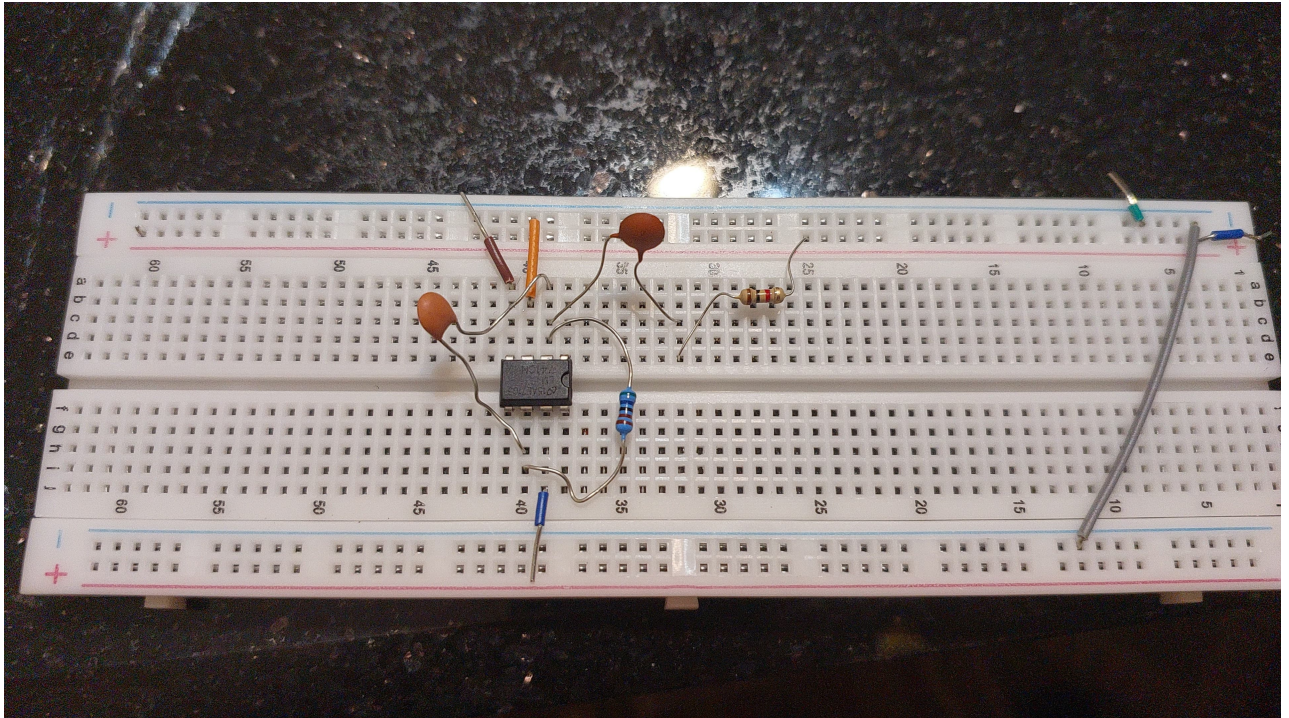


Figure 8: Close-up view of the hardware implementation on the breadboard.

Hardware results:

To verify the physical operation of the filter, a function generator was used to supply a sinusoidal input signal (V_{in}), and a digital oscilloscope was used to simultaneously measure V_{in} (Channel 1) and the output voltage V_{out} (Channel 2).

The circuit was tested at three distinct frequencies relative to the calculated center frequency ($f_0 \approx 3.45$ kHz) to observe the attenuation and amplification characteristics. Because this is an *inverting* band-pass filter, a 180° phase shift between the input and output signals is recognized.

i. $f < f_0$ ($f = 1 \text{ kHz}$)

At frequencies below the center frequency, the series input capacitor (C_1) exhibits high impedance, dominating the circuit and attenuating the input signal. The oscilloscope capture in Figure 9 demonstrates this high-pass behavior, where the amplitude of V_{out} is noticeably reduced compared to the expected maximum gain (gain is approximately $2 < 3.8$).

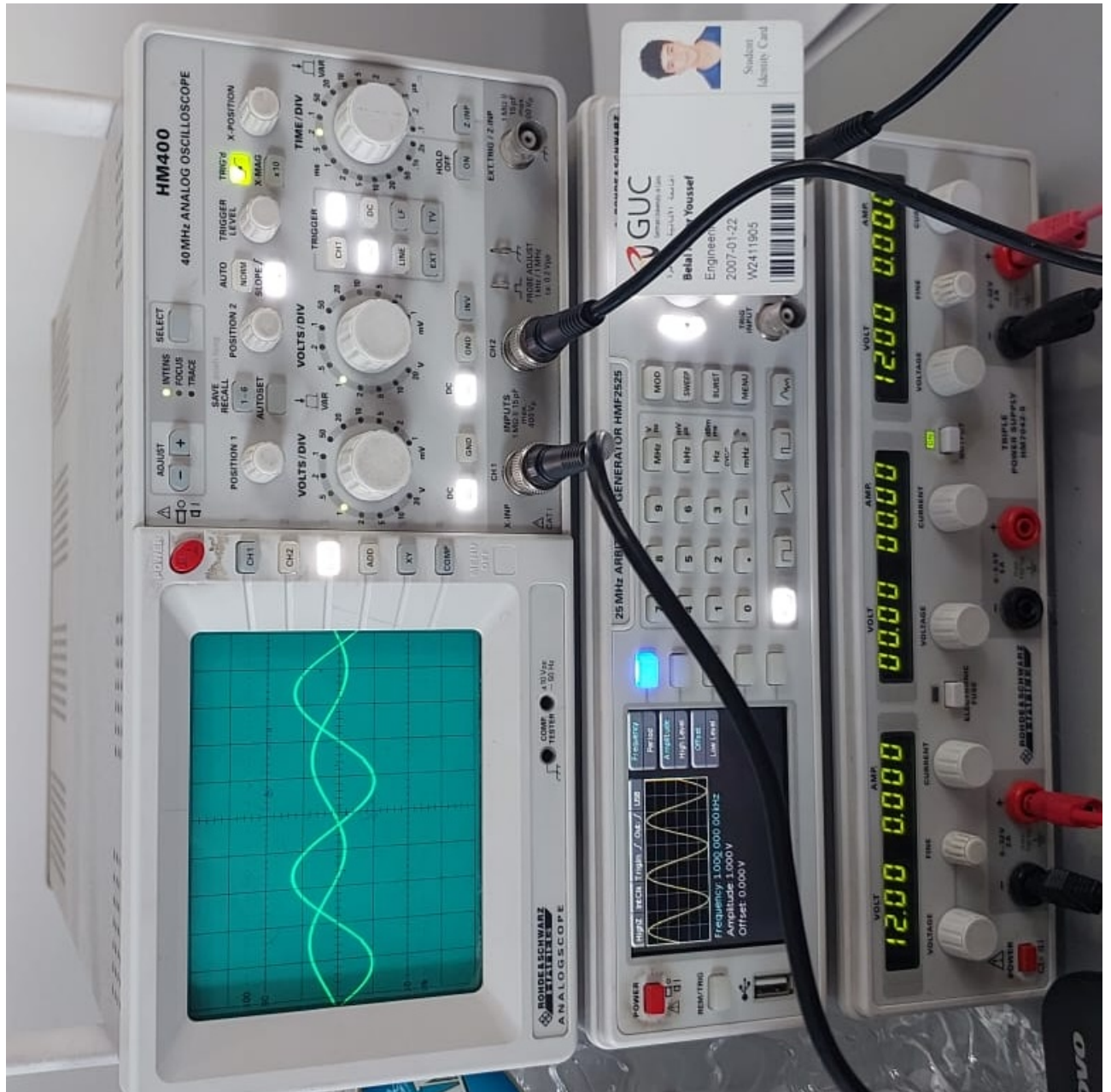


Figure 9: Oscilloscope capture at $f < f_0$. The output signal (CH2) shows attenuation.

ii. $f = f_0$ ($f = 3.45 \text{ kHz}$)

At the center frequency, the reactive components balance, allowing the circuit to achieve its maximum voltage gain ($|H_{\max}|$). As seen in Figure 10, the output voltage reaches its peak amplitude. The measured ratio of V_{out} to V_{in} is approximately 3.2 which closely aligns with the theoretically calculated maximum gain of 3.83. The expected 180° phase inversion is also clearly visible.

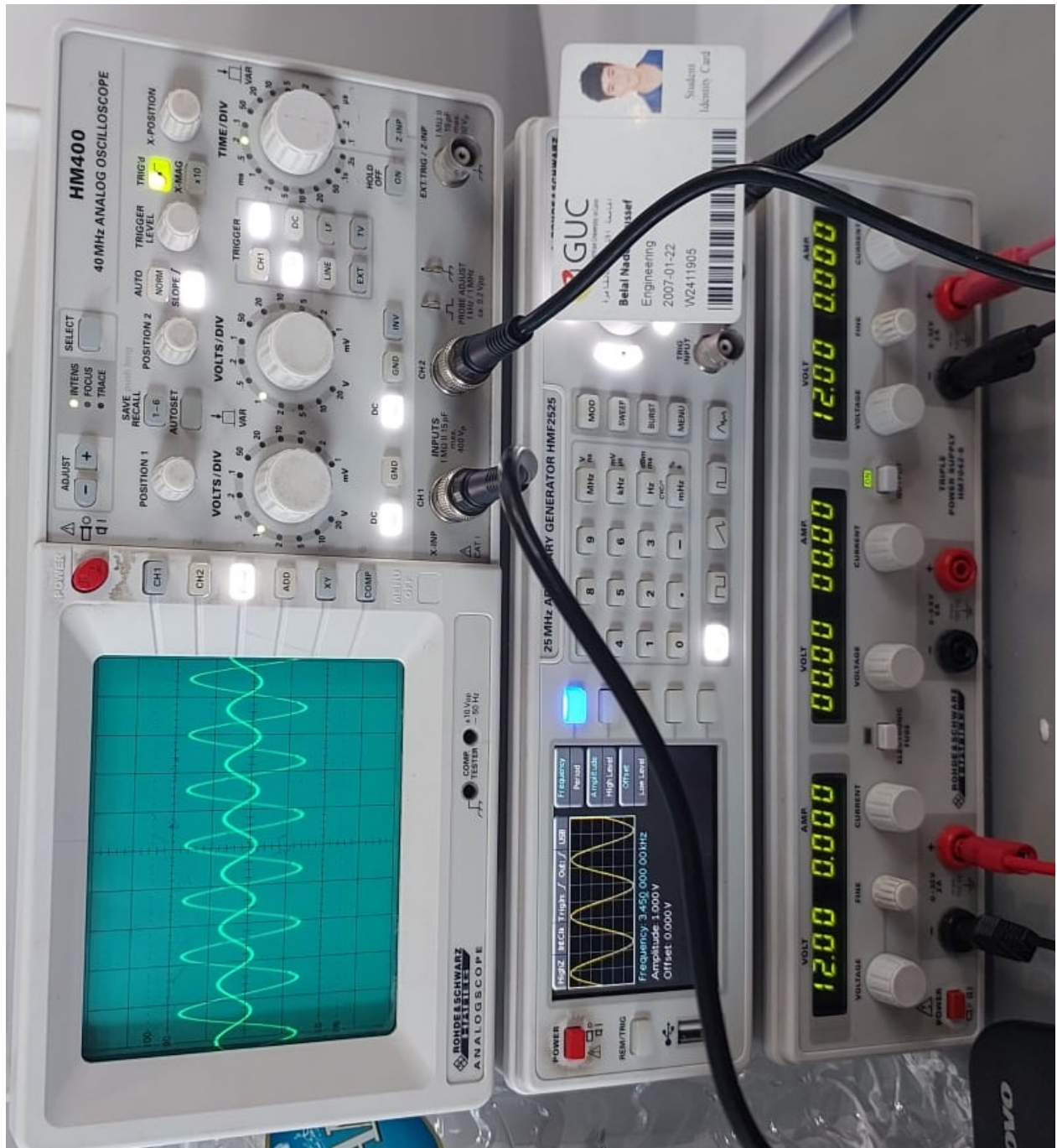


Figure 10: Oscilloscope capture at $f = f_0$. The circuit achieves maximum voltage gain.

iii. $f > f_0$ ($f = 10 \text{ kHz}$)

At frequencies above the center frequency, the parallel feedback capacitor (C_2) acts almost as a short circuit as the frequency increases, severely reducing the feedback impedance and pulling the output voltage toward the virtual ground. Figure 11 illustrates this low-pass effect, showing significant attenuation of the output signal once again.

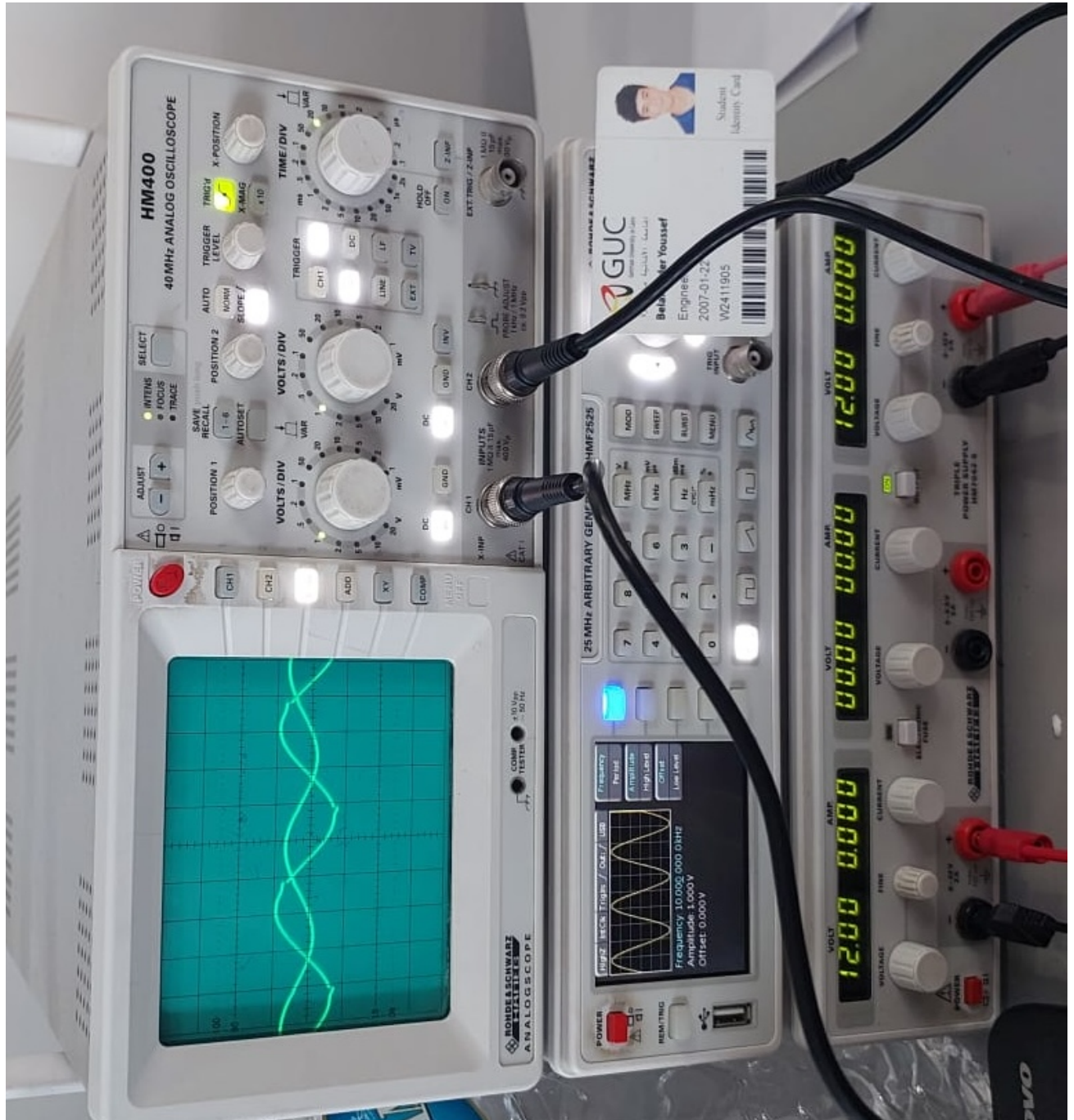


Figure 11: Oscilloscope capture at $f > f_0$. The output signal (CH2) is heavily attenuated.

Discussion of Results

The hardware implementation successfully validates the theoretical design and PSpice simulations. By observing the amplitude of V_{out} across the frequency spectrum, it is evident that the circuit allows a specific band of frequencies to pass with amplification (peaking at $f_0 = 3.45$ kHz) while attenuating frequencies outside this range. This behavior, coupled with the observed phase inversion, conclusively demonstrates that the constructed circuit functions as an Inverting Active Band-Pass Filter.